

Two regimes of bioinspired memory: an architecture-search perspective

Anonymous — double-blind review

Abstract—A robot learning over its service life must decide what to keep in a fixed-size memory. The episodic buffer of e-MDB, a cognitive architecture for lifelong open-ended robot learning, keeps the most recent episodes whatever their value. We present EMBER (Emergent Memory-Based Encoding and Reactivation), which casts such a memory as a six-axis design space—how a trace is read, written, weighted by surprise, decayed, evicted and reinforced—and evaluates all 576 configurations of one engine, among them a nearest-neighbour proxy of the e-MDB buffer, on how many importance-tagged experiences survive a stream that overruns capacity. The memory never sees the tags, only a surprise signal. The best configuration differs from that proxy on two axes, surprise-weighted encoding and weakest-trace eviction, and from the proxy neither changes anything alone. On a separable surprise signal it scores 0.818 against 0.106 and retains 0.983 of the tagged experiences against 0.050. It keeps the top rank in 13 of 14 held-out stream families; when recurring experience fits in memory it still retains 0.60 against the proxy’s 0.03 with surprise no better than chance; no external replacement rule beats it; and the buffer’s actual sequential read scores below the proxy. In MiniGrid, with surprise taken from reward prediction, it retains 47.2 % of rewarded events against 0.7 %. Which mechanism matters depends on the traces recurring experience occupies after consolidation relative to capacity: at ratios of 0.5 or less the write axis explains 73.0 % of the variance, above one 1.6 %; we state this as a hypothesis. On real data, merging needs repeated encounters to be separable from distinct ones, which CIFAR-100 embeddings are not. Better retention is not yet better behaviour: an episodic controller on the frontier integrates less return than on FIFO.

Index Terms—bioinspired memory, neural architecture search, Sparse Distributed Memory, engram consolidation, episodic memory, lifelong learning

I. INTRODUCTION

A cognitive robot that learns over its lifetime must decide what is worth keeping, keep it accessible in proportion to its relevance, and retrieve it from a partial or degraded cue. These requirements map onto three well-studied properties of biological memory: selective encoding modulated by surprise, use-regulated consolidation and decay, and content-addressable retrieval that is robust to noise.

The episodic buffer of e-MDB (*epistemic Multilevel Darwinist Brain*) [1], [2]—in its public ROS2 implementation, a fixed-capacity FIFO (first-in-first-out) queue [3]—satisfies none of the three. It keeps whatever was perceived most recently, reads episodes back by position or as training batches, and discards the oldest when full, whatever its importance. The bioinspired-memory literature offers principled remedies: Sparse Distributed Memory (SDM) [4] for noise-robust content-addressable retrieval; Hebbian engram consolidation [5], [6]; prediction-error gating of encoding, grounded in

dopaminergic modulation of hippocampal encoding [7], [8] and in the finding that larger reward-prediction errors improve episodic memory [9]; and eviction by value rather than age, since memory availability tracks the probability of future need [10] and forgetting is regulated in the service of decisions [11]. But these mechanisms are usually evaluated in isolation, inside architectures built to show them off, against baselines that do not share their design space, so a reported gain may come from the protocol rather than the mechanism.

EMBER (Emergent Memory-Based Encoding and Reactivation) inverts that approach. We define a discrete design space in which each mechanism is an axis, and traverse it exhaustively. A *configuration* is one choice on each of six axes; all 576 are enumerated, not sampled. The e-MDB buffer—more precisely, a FIFO-NN proxy of it that adds nearest-neighbour retrieval—is one point inside that space, which turns a potentially circular comparison into a falsifiable question: where does this *incumbent* land, and which configuration is the *frontier*, the best-scoring point? Each configuration receives a *surprise signal* with every experience: *novelty*, which the memory computes as one minus the cosine similarity to its closest trace, or *prediction error* in $[0, 1]$ from a model of what was expected. It is scored on *rare-event retention*, the fraction of importance-tagged experiences still recoverable after a stream many times longer than the capacity; the tag is used only for scoring, never shown to the memory. We relate the results to $r = K_{\text{proto}}/C$: the number of *prototypes* (distinct recurring kinds of experience) over the *capacity*, the number of traces held before a write forces an eviction.

Our contributions are:

- An exhaustive evaluation in which the e-MDB buffer is a point of the space. The frontier differs from the FIFO-NN proxy on two axes only, surprise-weighted encoding and weakest-trace eviction, and from the proxy neither changes anything alone. It outscores the proxy on a separable surprise signal, keeps that lead under degraded surprise and on held-out stream families, and is not beaten by seven memory-management rules from outside the space; the buffer’s real sequential read scores lower still.
- A *regime hypothesis*: which mechanism family governs retention is set by the ratio of the traces recurring experience occupies after consolidation to capacity, r_{eff} , which on clean streams equals r ; the gating mechanism the literature most motivates has a negligible main effect but a large one conditional on the eviction rule that reads it.
- A two-phase benchmark of four memory substrates that

admits them by reconstruction before testing retention under pressure: a pure spiking circuit fails on continuous vectors, and SDM’s retention comes from gated eviction with exact erasure, not from superposition.

- Two preconditions that appear only on real data: consolidation needs repeated encounters separable from distinct ones, which CIFAR-100 [12] embeddings are not; and in MiniGrid [13] surprise must be aligned with what defines importance, which, when importance is reward, raises the frontier’s retention from the floor to about half.

Section VIII collects, in one place, what these results do not show.

II. BACKGROUND AND RELATED WORK

Memory substrates. SDM [4] distributes each pattern over many overlapping *hard locations* with fixed random addresses, so a noisy query still activates enough of the same ones to reconstruct the original, and accuracy degrades gradually. Our instantiation draws 512 hard locations as random unit vectors; a write adds $s \cdot k$, with s the encoding strength, to the counters of the 25 (5%) most aligned with key k , and a read sums the counters of the query’s 25 nearest and normalizes. Activation is by inner product rather than a Hamming radius, so the radius adapts to local density (cf. [14]), and the memory keeps an explicit trace list alongside the counters, returning the stored trace most similar to the reconstruction. The *engram*, the physical trace of a memory [15], [5], rests on long-term potentiation (LTP): neurons that fire together strengthen their synapse [6]. The *Engram Neural Network* (ENN) is our name—not a prior architecture—for the minimal online memory built on that rule: fast Hebbian weights updated without backpropagation, where each write creates a trace or reinforces one above a cosine threshold. Spike-timing-dependent plasticity (STDP) [16] applies the rule to individual spikes and groups neurons into assemblies; the project’s third proof of concept (PoC-3) verified on leaky integrate-and-fire (LIF) neurons that a partial cue recruits the rest of its assembly.

Search, replay and memory management. Neural architecture search [17] optimizes architectural decisions; memory-augmented networks [18], [19] fix the memory type and learn to use it. We instead make the memory mechanism the search variable over a fully enumerable space, so no result can be attributed to an approximate optimizer. Prioritized experience replay [20] revisits transitions by temporal-difference error, the principle behind prediction-error gating, and Neural Episodic Control [21] reads a dictionary by similarity, as SDM does; both fix those choices in advance. Outside biology, caches evict by recency or frequency [22], [23]; reservoir sampling keeps a uniform sample of a stream [24] and is a strong replay baseline in continual learning [25]; rehearsal methods select exemplars by herding [26], gradient diversity [27], or surprise, reward and coverage in lifelong reinforcement learning [28], where replay mitigates forgetting [29]. These methods optimize downstream learning under a fixed rule;

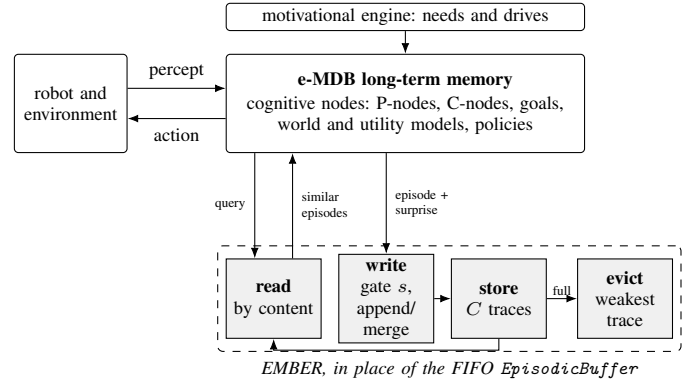


Fig. 1. Where EMBER would sit in e-MDB (a design; not implemented here). Each step yields an *Episode* that the public implementation appends to a *FIFO EpisodicBuffer*, from which world and utility models are trained [3]. EMBER would replace that buffer: *write* weights each episode by a surprise signal from e-MDB’s own models, *evict* discards the weakest trace when full, and *read* returns the stored episodes most similar to a query.

EMBER measures retention across a space of rules, and Section IV-C evaluates seven of them.

e-MDB and LOLA. e-MDB descends from the Multilevel Darwinist Brain [30] and targets LOLA (*Lifelong Open-ended Learning Autonomy*) [31]: a robot that acquires knowledge continuously, in domains unknown at design time and without predefined tasks. A long-term memory of *cognitive nodes*—perceptual classes, goals, contexts, world and utility models, policies—is activated from the current perception [32] and coupled to a motivational engine of needs and drives [33], [1], [2]. In the ROS2 implementation [3] each step yields an *Episode*, and the *EpisodicBuffer* from which world and utility models are trained is a bounded deque: episodes are appended, the oldest discarded, and read by index or as a batch, with no importance criterion and no similarity search (file `episodic_buffer.py`; no publication describes it). In our space its eviction and weighting are the configuration `{read: nn, write: append, strength: constant, decay: 1.0, evict: fifo, reinforce: 0}`; because that point retrieves by nearest neighbour, which the real buffer cannot, we call it the *FIFO-NN proxy* and treat it as an optimistic stand-in. Fig. 1 shows where EMBER would enter.

III. METHOD

A. The design space

Each axis is a decision an engineer building an episodic memory has to make anyway (Table I; Fig. 2 shows how they compose into one write). The *read* axis returns the single closest trace (`nn`), aggregates the three closest (`topk3`), or superposes all traces within a similarity radius of the best match (`radius`, the SDM read). The *write* axis always creates a trace (`append`) or first reinforces an existing one at cosine ≥ 0.85 (`merge`, Hebbian consolidation). The *strength* axis, the *salience gate*, assigns the encoding weight s . *Decay* erodes unreinforced strength; *evict* discards, when memory is

TABLE I
BIOINSPIRED DESIGN SPACE: 576 CONFIGURATIONS, EXHAUSTIVELY
ENUMERATED.

Axis	Options	Biological root
read	nn / topk3 / radius	Episodic retrieval vs. activation circle (Kanerva)
write	append / merge	Trace creation vs. Hebbian merge consolidation
strength	constant / novelty / pred_error / both	LTP gating by salience and surprise (dopamine)
decay	1.0 / 0.995 / 0.98	Synaptic weakening without reinforcement
evict	fifo / min_strength / min_utility / random	Forgetting by age vs. by importance
reinforce	0 / 0.5	LTP by reactivation

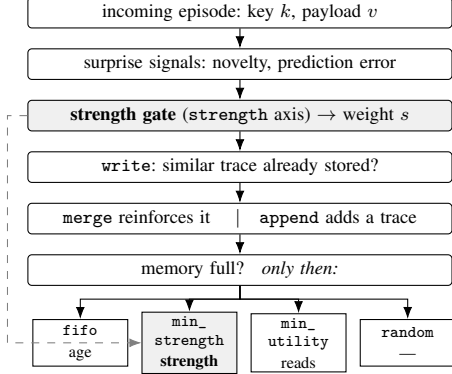


Fig. 2. The write path, and why the salience gate is conditional. The weight s is inert unless something downstream reads it; the dashed line marks the only consumer, `min_strength`, so in three quarters of the space the signal is computed and discarded. Eviction runs only when memory is full, which is why the regime transition of Section V falls where it does.

full, the oldest trace (`fifo`), the weakest (`min_strength`), the least read (`min_utility`, LFU-like) or a random one; `reinforce` raises a trace’s strength each time it is read. The 576 configurations are parameterizations of one engine in which every mechanism is implemented once, so any score difference is attributable to the axis settings alone. We reserve *architecture* for the four substrates of Section VI, which differ in how a trace is physically represented.

B. Experimental protocol

Every detail is taken from the code that produced the results. Keys live in $\mathbb{R}^{32}$ and are normalized at the memory boundary; every random draw derives from an explicit seed, and within a seed all configurations receive the identical stream.

Streams. K_{proto} centres are i.i.d. Gaussian vectors on the unit sphere. A routine experience picks a prototype uniformly, adds Gaussian noise ($\sigma = 0.05$ per dimension) and renormalizes; its payload is the prototype identity. Rare experiences have fresh random keys and unique payloads, at uniformly random positions. The robustness families of Section V each move one knob of this generator: Zipf visit frequencies, drift, intra-prototype noise, centre similarity, rare-event prevalence, recurrence, and C from 5 to 160. CIFAR-100 streams (Sec-

tion VII) use a different image of one of K_{proto} classes per visit and draw rare experiences from disjoint classes.

Surprise and importance. Prediction error is drawn from $\mathcal{N}(0.10, 0.05)$ clipped to $[0, 0.35]$ for routine experiences and from $\mathcal{N}(0.90, 0.05)$ clipped to $[0.65, 1]$ for rare ones, so the ranges do not overlap by construction. Novelty is one minus the highest cosine similarity to a stored trace (1 when empty). The gate assigns $s = 1$ (constant), $1 + 2\nu$ (novelty), $1 + 2e$ (pred_error) or $1 + 2\nu + 2e$ (both). The importance tag is the generator’s rare flag, used only for scoring.

Phase 1, reconstruction gate (capacity never binding). R1: 10 patterns, cues keeping 30/50/70% of the dimensions, scored by the cosine of the returned trace when its identity is correct and 0 otherwise. R2: 15 patterns under Gaussian noise $\sigma \in \{0.1, 0.2, 0.3, 0.4\}$, scored by identity accuracy. R3: capacity 2, write A then B , read A ; 10 pairs, scored as R1. R4: accuracy at $\sigma = 0.15$ for loads $\{2, 5, 10, 15, 20, 30\}$ with capacity equal to load. The gate is the mean of R1–R4; admission requires ≥ 0.50 .

Phase 2, battery ($C = 20$). T1, rare-event retention: $K_{proto} = 5$, 300 routine and 20 rare experiences; every 5 writes the memory is queried with an already written key, so reinforcement can act; at the end each rare experience is queried with its exact key and counts as retained if its payload returns at similarity ≥ 0.85 . T2, noisy retrieval: 60 random keys with prediction error $\sim \mathcal{U}(0.05, 0.95)$ in 20 slots; 120 queries pick an item with probability proportional to its prediction error and add noise $\sigma = 0.30$. T3, sequential interference: 4 blocks of 8 keys, each written 3 times, prediction error 0.9 in the first block and 0.1 after; the score is the fraction of first-block keys retained. The *aggregate score* is the unweighted mean of T1–T3, each averaged over seeds.

Seeds and statistics. The search uses 3 seeds per configuration; the threshold grid, 3 capacities $\times$ 8 ratios $\times$ 5 seeds on T1, with 20 visits per prototype and $\max(5, C/2)$ rare experiences; the substrate benchmark, 4 seeds; CIFAR-100, 5 ratios $\times$ 3 seeds at $C = 20$; MiniGrid, 40 rollouts per condition (200 in five seed blocks in the extended run). For an axis, $\eta^2 = SS_{between}/SS_{total}$ over the 576 configuration scores: a one-way, marginal decomposition. The factorial is full and balanced, so main effects are orthogonal; two-way interactions are cell-mean residuals (SS_{ab}/SS_{total}), reported separately. Scores are deterministic given the seed, so there is no error term and η^2 is descriptive. Bands are 95% percentile bootstraps (2000 resamples) with the seed as unit; a capacity’s crossover is the geometric mean of the last write-dominant and first eviction-dominant r .

Structural dependence. The axes are not functionally independent, and the decomposition does not assume they are. Strength is consumed only by `min_strength`, so `strength` and `reinforce` are inert in the three quarters of the space that never read it; `merge` acts only above its threshold; and under a constant gate with `append` writes and no reinforcement `min_strength` reduces exactly to `fifo`. A small main effect is therefore not evidence of a weak mechanism, and wherever a dependence is known we report the conditional effect. The

TABLE II
THE FIFO-NN PROXY OF THE E-MDB BUFFER AND THE FRONTIER,
WHICH DIFFER ON TWO AXES. NEITHER CHANGE WORKS ALONE. T1–T3
AS IN SECTION III-B.

Configuration	mean	T1	T2	T3
FIFO-NN proxy (incumbent)	0.106	0.050	0.269	0.000
+ gate or <code>min_strength</code> alone	bit-identical to the proxy			
+ both (frontier)	0.818	0.983	0.469	1.000

two-way shares locate it: `strength×evict` explains 4.3 % of the variance and `write×evict` 7.8 %.

IV. THE E-MDB PROXY AND THE FRONTIER

A. Where the proxy lands

Every axis is observable: a liveness audit, run in continuous integration, measures the largest score change between configurations differing only in that axis, and all six clear it. Write mode is the strongest main effect (56.6 %), followed by eviction (17.0 %), decay (1.8 %) and strength (1.4 %); read and reinforcement explain 0.0 % and 0.0 % as main effects despite positive observability (0.119 and 0.206).

The FIFO-NN proxy scores 0.106, rank 416–469 of 576 with 54 ties. It is not the floor: the configurations below it all write by append and combine minimum-utility eviction, topk3 reads, or `min_strength` with reinforcement. The frontier scores 0.818, almost eight times more.

B. Two axes, neither alone

The frontier is `{read: nn, write: append, strength: both, decay: 1.0, evict: min_strength, reinforce: 0}`: every experience is stored as its own trace with a weight proportional to its surprise, weights never decay or change on recall, and when memory fills the least surprising trace is discarded. It differs from the proxy on exactly two axes, and changing either alone leaves the proxy bit-identical (Table II); the salience gate alone is inert in every combination of the other axes. The reason is exact: under a constant gate with append and no reinforcement every trace has strength 1.0, so the minimum is always the oldest trace—`min_strength` is FIFO—and a gate feeding a policy that never reads strength computes a weight nothing consumes.

The same dependence explains why the gate’s 1.4 % main effect misleads. Within `{evict: min_strength, write: append}`, a constant gate retains 3.6 % of rare events and novelty plus prediction error 52.0 %, a factor of 14.41. Because routine and rare prediction errors do not overlap here, keeping the rare events follows almost directly from the construction once eviction reads strength: the factor shows that salience does nothing until something consumes it, not that selecting from a realistic signal is solved. The frontier never merges, so its mechanism is the same in both regimes of Section V; merging would accumulate strength on whatever recurs, until a prototype visited thirty times outweighs a rare event encoded once at maximum surprise.

C. How far the advantage holds

Degraded surprise (five seeds). With prediction error alone as the gate, the conditional factor falls from $16.27\times$ on the clean signal to $0.80\times$ (95 % CI 0.17–2.82) when the two ranges overlap completely; the combined gate keeps more only through novelty, itself a perfect classifier here, and falls to $3.10\times$ when a quarter of routine inputs are novel distractors and prediction error is at chance. Against the proxy, the frontier retains more rare events under every degradation at $r = 0.25$, still 0.60 against 0.03 at chance. Against the best salience-free configuration, which at $r = 0.25$ retains 0.76 by merging, its paired advantage holds only down to a surprise AUC of 0.95 under overlap, 0.96 under additive noise and 0.91 under 10 % inverted signals; it is reversed once half the surprise is credited one step late and at chance. At $r = 2$, where merging frees no space, it holds down to AUC 0.84.

Held-out streams. Re-scored on 14 held-out stream families with fresh seeds (Section V), the frontier keeps rank interval [1, 1] in all but one, [15, 20] with 0.1 % rare events, and the ranking of all configurations correlates with the standard one at Spearman ≥ 0.77 .

Rules from outside the space (T1–T3, five seeds), each replacing only the proxy’s eviction. Against the frontier’s 0.824, LRU, LFU and a frequency–recency utility cache stay at the proxy’s level (0.106, 0.096, 0.106); reservoir sampling, coverage selection [28] and stochastic prioritized replay reach 0.306, 0.439 and 0.500. Greedy surprise-prioritized eviction scores 0.789 (rank 4–5 of 577) because it coincides with an in-space configuration (prediction-error gate with `min_strength`); the frontier’s margin of 0.035 [0.028, 0.043] lies entirely in T3. What the frontier exploits is surprise read at eviction, a principle prioritized replay already uses, not a mechanism unique to bioinspired memory.

The buffer’s real read. The proxy is generous: the real buffer has no content-addressable read. We re-implemented its read as a scan in insertion order returning the first episode with cosine ≥ 0.85 —an assumption, since the buffer answers no queries. It keeps exactly the proxy’s traces but fails the reconstruction gate (0.368 against 0.903) and scores 0.010, last of 577; a match threshold of 0.5 restores the gate (0.826) but not the score (0.067). The proxy’s rank is thus an upper bound for the deployed buffer.

V. THE REGIME TRANSITION

The 56.6 % of the write axis is less a property of merging than of r . We sweep the space over capacities and ratios, holding visits per prototype fixed so that r is not confounded with recurrence. The attribution switches abruptly from the write axis to eviction near $r = 1$ (Fig. 3): the crossover, averaged over $C = 10, 20$ and 40, is 0.88 (descriptive interval [0.87, 0.89]), and the three capacities fall in the same grid interval. The mechanism is close to a restatement of what capacity means: eviction runs only when a write would exceed capacity, so below $r = 1$ recurring experience fits, merging compresses it and frees slots for the rare (the *compression* regime), while above it memory is saturated and the question

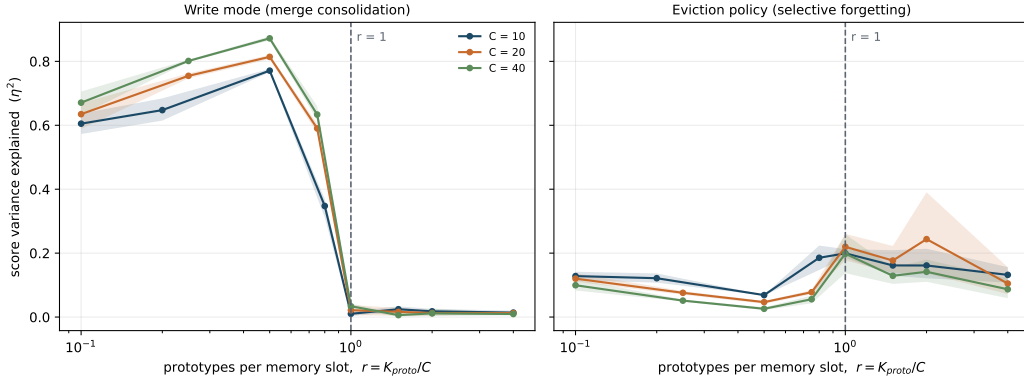


Fig. 3. Share of the variance in rare-event retention across the 576 configurations attributable to the write axis (left) and the eviction axis (right), against $r = K_{proto}/C$, for three capacities. If absolute capacity governed the transition the curves would cross at three different points; if stream redundancy did, fixing visits per prototype would have flattened the effect. Neither happens at this resolution. Bands: 95 % bootstrap over the five seed-level values.

becomes which policy preserves what matters (*selection*). What the grid adds is that the switch is abrupt, not that it exists. η^2 is direction-blind: it shows that the write axis moves the outcome, not that merging is preferable, and above the threshold write and eviction jointly explain only about a fifth of the variance, the rest belonging to the gate, decay and their interaction with eviction.

Nominal or effective ratio. Repeated on 23 stream families (Section III-B), nominal r is not the governing variable: its crossover ranges from 0.16 to 1.32 (s.d. of its log 0.46); drift and noise pull it far below 1, at the noisiest setting eviction dominates at every r , and even capacity moves it, from 0.71 at $C = 5$ to 0.98 at $C = 160$. Over $r_{eff} = K_{effective}/C$, where $K_{effective}$ counts the traces the routine occupies after consolidation, crossovers collapse to [0.56, 1.28] (s.d. 0.17), tighter than nominal r in 99.8 % of bootstrap replicates and than four other pressure measures fixed in advance; rare-event prevalence and $C = 5$ leave the residual spread. The hypothesis is therefore stated over r_{eff} , which equals r when encounters consolidate cleanly, as in Fig. 3.

Merge threshold. Sweeping it from 0.30 to 0.97 ($C = 20$, five seeds) leaves the nominal crossover at $r \approx 0.87$ for every threshold from 0.40 to 0.85; at 0.90 merging misses part of the visits and it drops to 0.61; at 0.30 merging absorbs 38.0 % of rare events; from 0.95 merging fragments prototypes and eviction dominates at every r . Over $K_{effective}$ the crossover stays between 0.86 and 1.02 for every threshold from 0.30 to 0.90, and a sweep of the gate’s gain (0.5–8) leaves the pattern unchanged.

VI. MEMORY SUBSTRATES

The search ranks configurations of one engine, and the frontier reads by nearest neighbour, which never exercises a distributed representation. We therefore compare four substrates—SDM, ENN, Spiking-SDM (SDM with LIF activation) and a pure spiking circuit (LIF with STDP)—with the FIFO-NN proxy (“FIFO”) as a point of the same space, admitting them by reconstruction (R1–R4) before testing them under pressure (T1–T3, Table III).

All but the pure spiking circuit clear the gate (0.459). It forms distinct assemblies (perfect $A \rightarrow B$ interference) but fails reconstruction and noise robustness: it was built for discrete activation patterns, and over a continuous vector inner-product neuron selection is sensitive to small perturbations, so a noisy cue activates a different set. This motivates EMBER’s PoC-5, pairing SDM for the continuous-to-distributed mapping with ENN for the trace lifecycle. ENN and FIFO tie at the gate: with no pressure the 0.85 merge threshold is rarely exceeded between near-orthogonal vectors, so both reduce to nearest-neighbour search.

Under pressure with the clean signal, SDM retains every rare event (1.000, $\sigma = 0.000$ over four seeds); ENN, with the same gate and eviction, retains 0.750, because merging accumulates strength on routine visits; Spiking-SDM retains 0.675, and the FIFO is the floor (0.038, $\sigma = 0.041$). On sequential interference SDM keeps 0.875 of the first block, ENN 0.375 and the FIFO 0.000. This phase has a ceiling: twenty rare items enter twenty slots with separable prediction errors, so perfect retention confirms correct wiring, and the informative comparison is SDM against ENN, which share the gate. (The threshold grid, with rare events at half the capacity, removes the exact-fit ceiling but not the separability.) Under degraded surprise (Section IV-C; retention precision equals recall here by arithmetic) SDM falls to 0.81 at AUC 0.99 and 0.60 at 0.95, below ENN, whose merging holds 0.75 under every prediction-error degradation; in the $r = 2$ stream, where precision is capped at 0.5 by ten rare items in twenty slots, SDM beats the FIFO down to AUC 0.74.

Cost. Capacity counts traces, not bytes: SDM’s 512 hard locations hold 41.96 times the FIFO’s bytes at $C = 20$ (Table III) and take 28 against 12 μ s per read. An ablation shows that retention needs gated eviction with exact erasure, not superposition: the trace list alone, read by nearest neighbour, keeps every rare event and scores 0.788 against 0.741 for the full SDM; counters alone keep them only if an evicted trace is subtracted exactly (1.000, else 0.005). SDM is stable in its own hyperparameters: the T1–T3 mean moves 0.091 from 64

TABLE III

TWO-PHASE EVALUATION. R1–R4 MEASURE RECONSTRUCTION WITH NO CAPACITY PRESSURE AND DECIDE ADMISSION; T1–T3 MEASURE RETENTION UNDER PRESSURE. KiB: STATE HELD AT $C = 20$, $d = 32$.

Arch.	R1	R2	R3	R4	Mean	Gate	T1	T2	T3	KiB
SDM	0.672	0.788	1.000	0.978	0.859	✓	1.000	0.365	0.875	131.1
ENN	0.691	0.938	1.000	1.000	0.907	✓	0.750	0.344	0.375	3.1
Spiking	0.190	0.146	1.000	0.499	0.459	×	–	–	–	105.1
SpikingSDM	0.507	0.433	0.950	0.588	0.619	✓	0.675	0.158	0.719	133.0
FIFO	0.691	0.938	1.000	1.000	0.907	✓	0.038	0.254	0.000	3.1

to 2048 hard locations; rare retention is perfect for activation fractions 0.005–0.2 and drops only at 0.5 (0.395), which, like an admission threshold above 0.5, fails the gate.

VII. REAL DATA

A. CIFAR-100: the separability precondition

The synthetic keys are nearly orthogonal Gaussian vectors, which could overstate retrieval. We rerun the grid on CIFAR-100 [12] embeddings from a pretrained ResNet-18 [34], reduced by principal component analysis to 32 dimensions. With this extractor the write axis is inoperative in all five cells, explaining 0.0–0.4 % of the variance whatever the ratio, while the synthetic domain in the same run crosses normally ($\hat{r} \approx 0.77$ over the estimated $\hat{K}/C$). The deciding quantity is *intra-prototype similarity*, how alike two encounters with one class look. Its median, 0.401, is far below the 0.85 threshold, so merging almost never fires and $K_{effective}$ grows from 84 to 1435 traces as nominal K rises from 5 to 80 at $C = 20$, instead of converging to K_{proto} . Lowering the threshold does not help, because intra-class similarity (median 0.38) overlaps inter-class similarity (95th percentile 0.42): no threshold from 0.30 to 0.97 yields merges both frequent and pure, and from 0.40 up eviction dominates on average. At 0.30 the write axis regains dominance at low r , but not as useful compression: only 47.7 % of merges join the same class, 70.0 % of rare events are absorbed on arrival, and the best merging configuration retains 0.30 against 1.00 for the append frontier.

The precondition is thus *separability* —a threshold that repeated encounters clear and distinct ones do not— and it is not a term in r but a check that comes first: where consolidation is partial, sub-threshold visits fragment a prototype and the hypothesis holds over r_{eff} ; where it is unreachable, $K_{effective}$ grows with the stream and no ratio, nominal or effective, produces a compression regime. Estimating K_{proto} does not rescue it either. We take the cluster count maximizing the mean silhouette coefficient, require a silhouette of at least 0.15 before treating r as defined (clustering returns an argmax even on isotropic noise), and report a bootstrap interval; on CIFAR-100 the intervals reach the ceiling of the search in several cells, consistent with no compact cluster structure.

B. MiniGrid: aligning surprise with importance

A robot does not receive a stream: it receives the consequences of its own actions, under partial observation. On MiniGrid-MemoryS13-v0 [13], where the agent must remember an object seen at the start to choose correctly at

the end, we compare the frontier and the FIFO on rare-event retention, with rewarded steps as the important events (40 rollouts per condition). Under a random policy, with surprise from the one-step prediction error of an online linear transition model, both sit at the floor (0.0 % and 2.6 %): 75.4 % of routine steps are at least as surprising as the least surprising of the 38 rewarded ones. A forward-biased policy quadruples the rewarded events (156) without changing this overlap (73.5 %) or the floor, so the problem is not sample size. The dopaminergic account that motivates the gate concerns *reward-prediction error* [35], [9]. Predicting reward instead, with the same model and policy, cuts the overlap to 5.6 % and lifts the frontier to 52.6 % (95 % CI by bootstrap over rollouts [37.9, 69.0] %; design effect 0.92), while the FIFO, which never reads strength, stays at 2.6 % ([0.0, 8.6] %). Over 200 rollouts in five seed blocks the frontier retains 47.2 % ([39.0, 56.3] %) against 0.7 % for the FIFO, and the same frontier fed a uniform random signal retains 0.7 %: the gain comes from the signal, not from eviction by strength.

Against that null control we cross four MiniGrid environments, four policies (random, forward-biased, noisy planner, tabular Q-learning), five definitions of importance and five candidate signals, 350 cells. The gain tracks how well the signal separates what matters (Spearman $\rho = 0.67$ against its AUC), averaging 37.3 points where $AUC \geq 0.97$ and 1.8 near 0.5. Count-based novelty transfers when importance is novelty (it beats the null in 16 of 16 cells). The salience mechanism thus transfers when surprise is anchored in the variable that defines importance; Section VIII returns to the cases where no causal signal we tried does.

VIII. DISCUSSION AND LIMITATIONS

What holds against e-MDB. On the synthetic battery, with a separable signal, the frontier outscores the FIFO-NN proxy by a factor of almost eight and retains nearly every rare event where the proxy keeps one in twenty; the lead over the proxy survives 13 of 14 held-out stream families and, at $r = 0.25$, every surprise degradation we tried; no external rule beats the frontier; and the buffer’s real sequential read, under our assumed match rule, scores lower still. In MiniGrid, with a signal aligned with importance, it retains about half of the important events against almost none. This is not a criticism of e-MDB, whose buffer feeds model training rather than recall, but a specific claim: as a store of what to remember, the buffer as implemented is the floor at rare-event retention under capacity pressure. Given a surprise signal that separates what

matters, the recommendation does not require a robot to know its own r_{eff} , since the frontier never merges.

What does not. Four results bound that claim. First, *retaining better is not yet acting better*. An episodic controller [21] whose only experience is the memory faced four MiniGrid goal variants that alternate and recur (ten seeds). At $C = 300$ the frontier earns more than FIFO on the first episodes of a returning task (+0.038, 95 % CI [0.005, 0.071]) but integrates less return over the stream (area under the curve -0.075 , $[-0.098, -0.053]$, no seed favouring it), and likewise at $C = 1000$: without decay, retention costs plasticity. Decay (0.995 or 0.98) lifts its area to the FIFO level, and its return-task advantage shrinks to within noise of FIFO (lost at $C = 1000$): no tested configuration is significantly better than FIFO on both. The frontier does beat reservoir sampling on return, and an unbounded memory (0.566 against 0.144) shows how far every bounded policy remains from the ceiling. Second, *at equal bytes the FIFO wins*: at SDM’s budget (144 KiB) a FIFO holds 921 traces, more than T1 writes, so no capacity pressure is left, and scores 0.924 against SDM’s 0.745: SDM’s footprint buys nothing a FIFO of equal bytes cannot match, and the advantage measured here is one of selection under capacity pressure. Third, *the advantage depends on the quality of the surprise signal*: the factor of 14.41 belongs to a separable signal, and at $r = 0.25$ a surprise AUC below about 0.95 hands the advantage to a configuration that merges. In MiniGrid perceptual prediction error leaves retention at the floor, and when importance is delayed no causal signal reliably retains the steps that precede a reward: reward-prediction error beats the null in 0 of 15 cells, TD error in 2 and count novelty in 6, with a mean gain of 1.8 points, while only the non-causal discounted return does (15); when importance is a key event, not even the return beats the null (0 of 8). Fourth, *the regime is set by r_{eff} , not nominal r* : drift or noisy encounters fragment one recurring situation into several traces and push a robot into selection earlier than counting kinds of situation suggests.

The open tension. LOLA is defined by the absence of predefined goals, and an open-ended robot may face long stretches with no reward to predict; the available signal then falls back to perceptual prediction error, where we measured retention at the floor. The fix is valid where a reward exists and unproven where it does not, the regime e-MDB is built for. Learning progress or the needs and drives of e-MDB’s motivational engine [33] are testable substitutes that we do not evaluate.

Untested. EMBER is not integrated into the running e-MDB: Fig. 1 is a design, the buffer’s read was re-implemented rather than run, and whether better retention improves the world and utility models e-MDB trains from its buffer is untested. The separability check was measured with class labels; a label-free online estimate is untested. The real-data evidence uses one visual extractor (ImageNet-pretrained ResNet-18, PCA to 32 dimensions), one dataset and one capacity, so whether the low intra-class similarity belongs to the domain or the extractor is open; a fine-tuned extractor could change it, a lower merge threshold does not. MiniGrid

spans four environments and four policies but stays within small discrete grid worlds, synthetic drift enters only the robustness families, and no experiment ran on a physical robot, so real sensor noise, continuous control and lifelong non-stationarity remain untested.

IX. CONCLUSION

Treating the e-MDB episodic buffer as one point of an exhaustively enumerated design space shows what it lacks and what fixes it. Two axes separate it from the frontier — surprise-weighted encoding and weakest-trace eviction— and, starting from the buffer’s proxy, neither does anything without the other. Together they lift the aggregate score from 0.106 to 0.818 and rare-event retention from 0.050 to 0.983 on a separable signal; the lead over the proxy holds on held-out stream families and, when recurring experience fits in memory, even at chance-level surprise; no external rule beats it; the buffer’s real read scores lower still, and in MiniGrid the frontier retains 47.2 % of rewarded events against 0.7 %. Which mechanism matters is set by the regime, located by r_{eff} ; SDM’s retention comes from gated eviction with exact erasure; a pure spiking circuit needs an encoding layer for continuous vectors; and consolidation and salience each carry a precondition —separable encounters, and a surprise signal aligned with importance— that only real data reveal. The claim is about retention under capacity pressure: whether it becomes better behaviour, and how to define salience without reward, are the questions that remain.

CODE AND DATA AVAILABILITY

The code and configuration files for the design space, tasks and all experiments, with the measured results (results/**/*.*.json, with a provenance manifest per experiment) behind every number in this paper, will be released in a public repository under an open license.

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